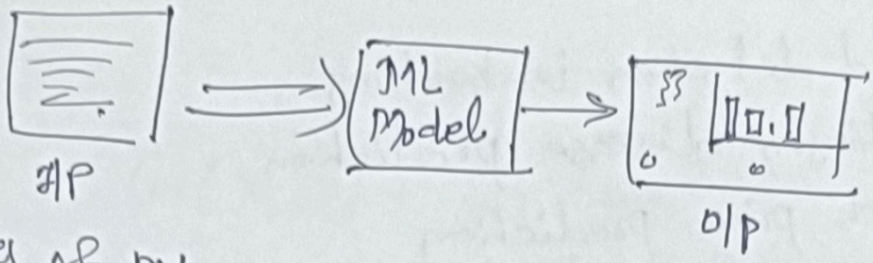
Q.1 A

Soln →

What is m/c learning? Explain different types of m/c learning algorithms. For each type discuss any two reals life applications.

M/c learning is a branch of AI that enables algorithms to uncover hidden pattern within dataset. It allows how to predict news, similar data without explicit programming for each task.



Types of ML

1. Supervised learning
It is about the training of labeled datased to map into features to target.
Ex. Regression → predicts continuous values.
Classification → Assigns data to sp. categories

2. Unsupervised m/c learning Algorithms.
It identifies patterns in unlabelled data.
key types are
- Clustering - groups similar data points
- dimensionality reduction → It reduces the number of variables while learning key variables or key information.

3. Reinforcement ML learning
In this the agent learns by interacting

with an environment, performing actions, and receiving rewards/penalties

There are two types of reinforcement learning

① Model based reinforcement learning
Here model learns from the environment & plans activities

② The agent learns from the ~~environment~~ experience without a model

Applications.

1) Fraud detection in banking
2) Parkinson disease prediction
3) Stock price prediction
4) Cancer cell classification.
5) Customer segmentation.
6) Recommendation system
7) Social N/w Analysis.

Q.1 B Some first Principal component & corresponding data transformation values.

Ip Data:

Customer	X1	X2
A	4	2
B	2	3
C	3	4
D	3	6

Step 1:

Data matrix $\begin{bmatrix} 4 & 2 \\ 2 & 3 \\ 3 & 4 \\ 3 & 6 \end{bmatrix}$

Step2 Std. the data

PCA works best when the data is centered (zero mean). We substract the mean from each feature

$\mu_1 = \frac{4+2+3+5}{4} = 3.5$

$\mu_2 = \frac{2+3+4+6}{4} = 3.75$

Standardize matrix.

$X_{std} = X - \mu$: $X_{std} = \begin{bmatrix} 0.5 & -1.75 \\ -1.5 & -0.75 \\ -0.5 & 0.25 \\ 1.5 & 2.25 \end{bmatrix}$ ✓

Step3: Compute Covariance matrix

$$Cov(X_{std}) = \frac{1}{n-1} X_{std}^T X_{std}$$

$$= \frac{1}{3} \begin{bmatrix} 0.5 & -1.5 & -0.5 & 1.5 \\ 1.75 & -0.75 & 0.25 & 2.25 \end{bmatrix} \begin{bmatrix} 0.5 & -1.75 \\ -1.5 & -0.75 \\ -0.5 & 0.25 \\ 1.5 & 2.25 \end{bmatrix}$$

$$= \begin{bmatrix} 1.6667 & 2.1667 \\ 2.1667 & 3.5833 \end{bmatrix}$$

Step4 Compute Eigenvalues & Eigenvectors.

$Cov(X_{std}) \cdot V = \lambda V$

$$\begin{bmatrix} 1.6667 & 2.1667 \\ 2.1667 & 3.5833 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \lambda \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

We get $\lambda = 5.19$, $\lambda_2 = 0.06$

∴ Eigenvectors principal components.

$PC1 = \begin{bmatrix} 0.52 \\ 0.85 \end{bmatrix}$, $PC2 = \begin{bmatrix} -0.85 \\ 0.52 \end{bmatrix}$

Step5 - Project the data onto Principal component (PC1) for 1-D projection.

∴ $Z = X_{std} \cdot PC1 = \begin{bmatrix} -1.14 \\ -1.46 \\ -0.16 \end{bmatrix}$ ✓

Q.2 A) Dimensionality redn techniques

- It helps in reducing the feature space while retaining imp. infn

Two types of dimensionality reduction techniques exists.

① Feature selection - choosing subset of the original features, keeping only relevant

② Feature extraction - Creates new features from original one, combining them in a way that captures the most important infn in fewer dimension

Common dimensionality redn techniques are

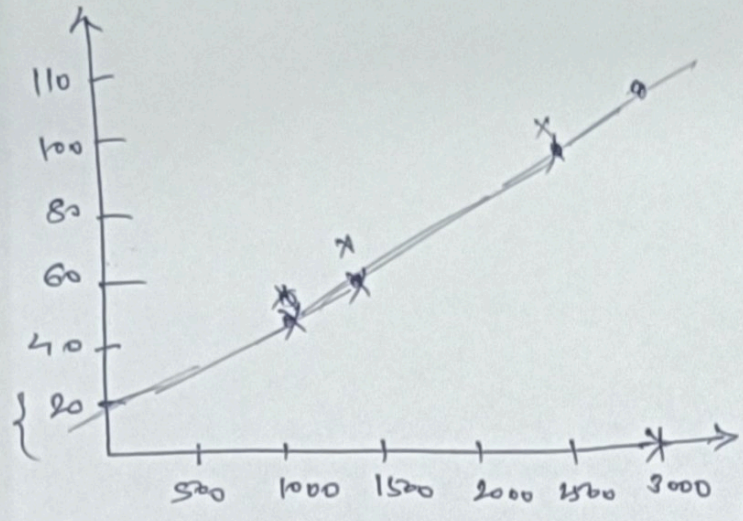
- PCA - It identifies the principal components, which are orthogonal directions of max variance in the data. It projects data into lower dimension subspace.

- LDA - Linear technique used for dimensionality reduction & classification, aiming to find lower dimensional space that maximizes class separability.

Q.LB Linear regression.

Use the following data to predict the price of a house i.e. 2200 sq.ft in size

Size (sq ft)	Price
1000	50
1500	65
2000	80
2500	95
3000	110



Price = m. Size + b.

$m = \frac{y_2 - y_1}{x_2 - x_1} = 0.03$, $b = 20$.

We have our eq

$Y = b_0 + b_1 X + e$.

(Assume $e = 0$)

Price = $20 + 0.03 \times Size$

$= 20 + 0.03 \times 2200$

Price = 86

∴ The predicted price for 22 sq.ft house is 84L.